

IEORE4004

Optimization Models and Methods

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Outline

- Nonlinear Programming
 - Introduction to Nonlinear Programming
 - Examples of Nonlinear Programming
- Convexity
- Quadratic Programming
 - Introduction to Quadratic Programming
 - Quadratic Programming Using Gurobi

Nonlinear Programming

- Introduction to Nonlinear Programming
- Examples of Nonlinear Programming

Introduction to Nonlinear Programming

Introduction To Nonlinear Programming (NLP)

Given:

$$\begin{aligned} \min \quad & f(x_1, x_2, \dots, x_n) \\ & g_i(x_1, \dots, x_n) \leq b_i, \quad i = 1, \dots, m \\ & h_i(x_1, \dots, x_n) = b'_i, \quad i = 1, \dots, m' \end{aligned}$$

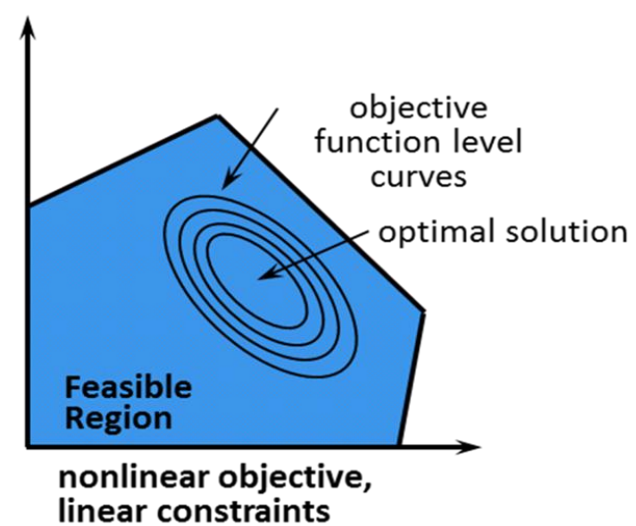
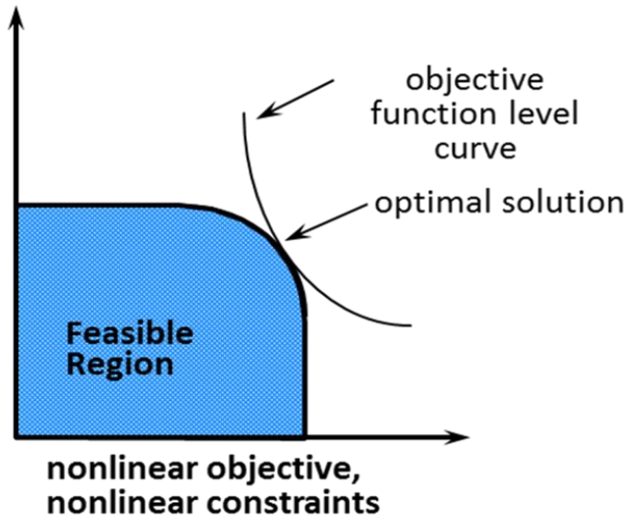
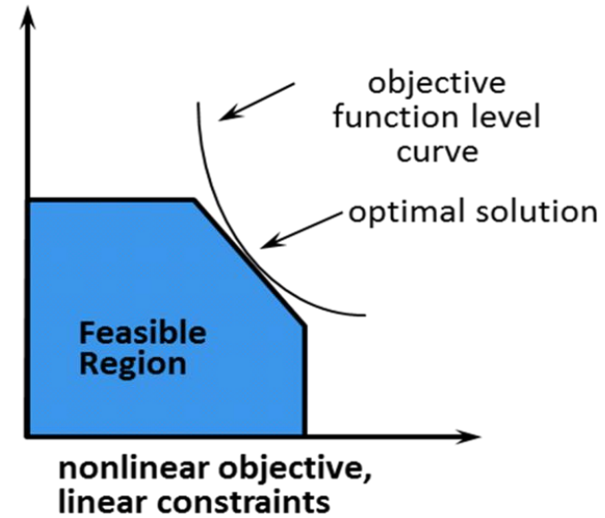
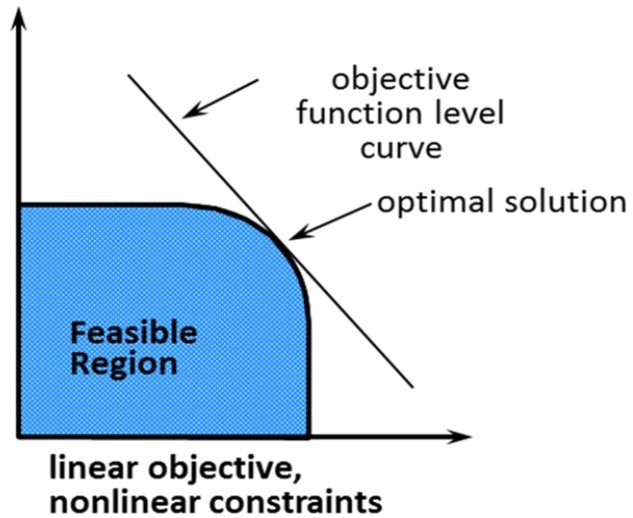
This is an NLP if at least one of $f(x)$, $g(x)$, or $h(x)$ is nonlinear

Methods: There is no generalized method that works for all NLPs!

The optimal solution can be found anywhere; it may exist:

- on the boundary of the feasible region,
- or at an interior point

Some Possibilities for NLPs...



Graphic thanks to *A Practical Introduction to Management Science*, by Cliff Ragsdale, 6th edition

Real-world Is Nonlinear In Nature...

Nonlinear optimization problems occur in a wide variety of domains

- Natural phenomena exhibiting *nonlinear* behavior (gravity, energy, ...)
- Manufacturing
 - *Materials properties*
- Chemical processes
 - *Blending, cracking, ...*
- Financial
 - *Economies of scale, risk measures, ...*
- Many, many more...

A Portfolio Optimization Problem

An investor has \$5,000 and two potential investments. From historical data, investments 1 and 2 have an expected annual return of 20 and 16 percent, respectively. How should she invest to maximize her return?

$$\text{Max} \quad f(x) = 20x_1 + 16x_2$$

$$\text{subject to:} \quad x_1 + x_2 \leq 5$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

A Portfolio Optimization Problem

Now suppose that there is some risk associated with those investments, and that the variance of the total return is:

$$2x_1^2 + x_2^2 + (x_1 + x_2)^2$$

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$$\begin{aligned} \text{Max} \quad & f(x) = 20x_1 + 16x_2 - \theta[2x_1^2 + x_2^2 + (x_1 + x_2)^2] \\ \text{subject to:} \quad & x_1 + x_2 \leq 5 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}$$

A Portfolio Optimization Problem

Now suppose that there is some risk associated with those investments, and that the variance of the total return is:

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Max
subject to:

$$f(x) = 20x_1 + 16x_2 - \theta[2x_1^2 + x_2^2 + (x_1 + x_2)^2]$$

$$x_1 + x_2 \leq 5$$

$$x_1 \geq 0$$

$$x_2 \geq 0$$

$\theta \geq 0$, risk averse
 $\theta < 0$, risk seeking

Recall Linear Regression

We are given vectors

We are given scalar values

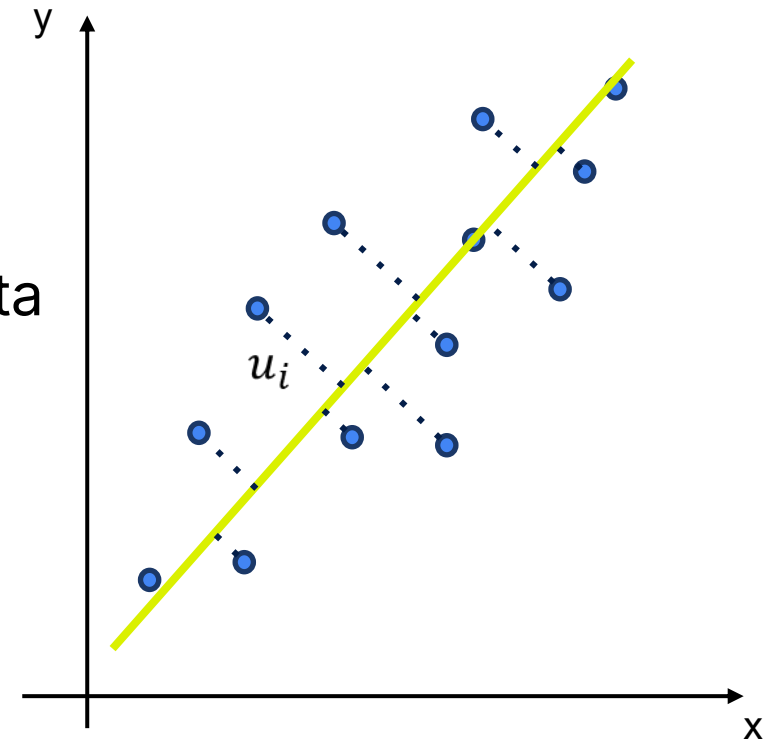
We want to “approximate” a linear model for the data

We want to find vectors $w \in \mathbb{R}^n, z \in \mathbb{R}$, so that:

$$y_i \approx w^T x_i + z, \quad 1 \leq i \leq N$$

For a given choice of w, z , the error on (x_i, y_i) is:

$$u_i = y_i - (w^T x_i + z)$$



Recall Linear Regression

We are given vectors

We are given scalar values

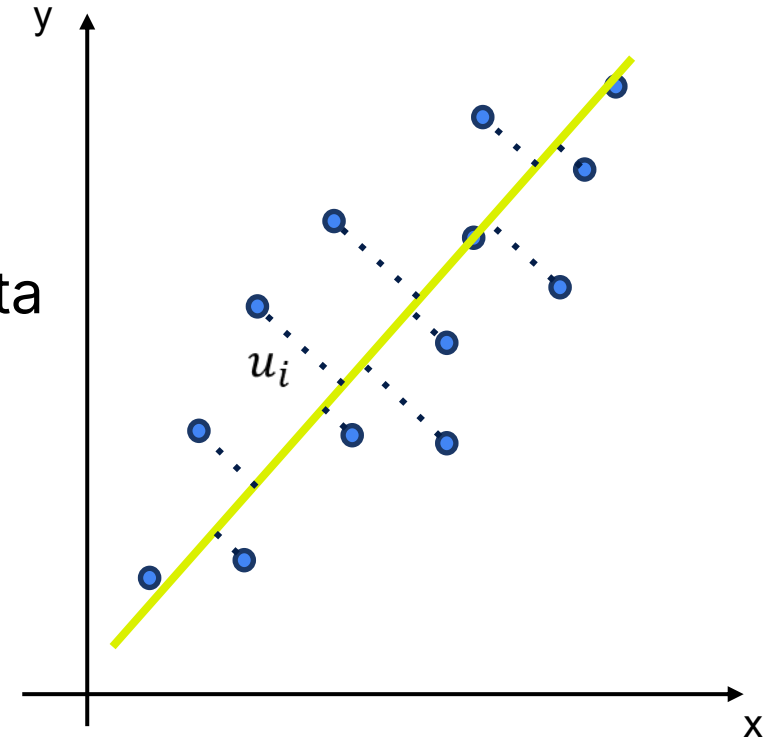
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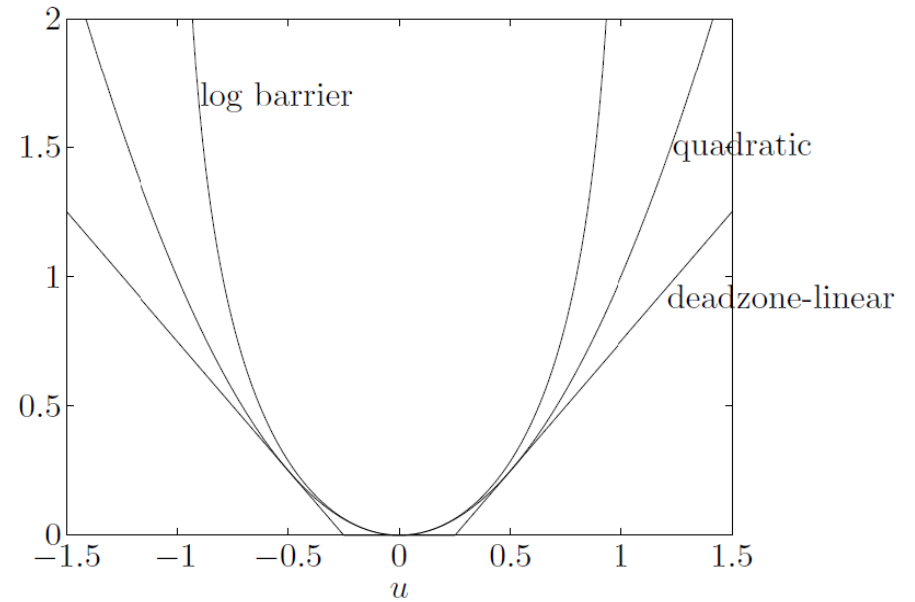


A linear regression model solves $\min_{w,z} \ell(u)$ for an appropriate function ℓ

Linear Regression Models

Some common choices for l :

- **Quadratic** loss: $l(u) = \sum_{i=1}^N u_i^2 =: u^2$
- **Absolute** loss or L_1 loss: $l(u) = \sum_i |u_i| = |u|$
- **Deadzone** linear: $l(u) = \begin{cases} 0, & |u| < 1 \\ 2|u| - 1, & \text{otherwise} \end{cases}$
- **Huber** penalty: $l(u) = \begin{cases} u^2, & |u| < 1 \\ 2|u| - 1, & \text{otherwise} \end{cases}$
- **Log-barrier**: $l(u) = -\log(1 - \frac{u^2}{M})$ for some constant M



Examples of Nonlinear Programming

NLP Examples

1. Maximize $f(x) = x_1^3 + x_2^2 + 2x_2$

Without constraints

NLP Examples

1. Maximize $f(x) = x_1^3 + x_2^2 + 2x_2$

2. Maximize $f(x) = x_1^2 + x_2^2$,

subject to $x_1 + x_2 \geq 4$,

$2x_1 + x_2 \geq 5$,

$x_1, x_2 \geq 0$

Without constraints

With constraints
(inequality)

NLP Examples

1. Maximize $f(x) = x_1^3 + x_2^2 + 2x_2$

Without constraints

2. Maximize $f(x) = x_1^2 + x_2^2$,

subject to $x_1 + x_2 \geq 4$,

$$2x_1 + x_2 \geq 5,$$

$$x_1, x_2 \geq 0$$

With constraints
(inequality)

3. Minimize $f(x) = 2x_1^2 - 24x_1 + 2x_2^2 - 8x_2 + 2x_3^2 - 12x_3 + 200$

subject to $x_1 + x_2 + x_3 = 11$,

$$x_1, x_2, x_3 \geq 0$$

With constraints
(equality constraints)

NLP Examples

4. Minimize $f(x) = x_1^2 + x_2^2 + 5x_1x_2$

Without constraints

Quadratic form

NLP Examples

4. Minimize $f(x) = x_1^2 + x_2^2 + 5x_1x_2$

Without constraints

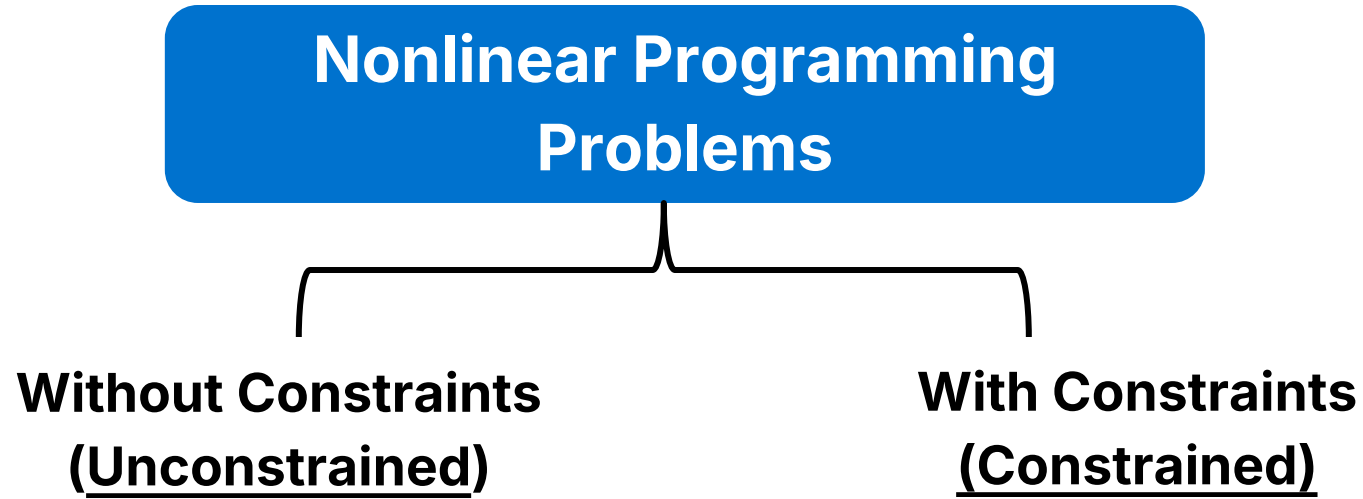
Quadratic form

5. Minimize $f(x, y) = x^3 + 5e^{-y}$

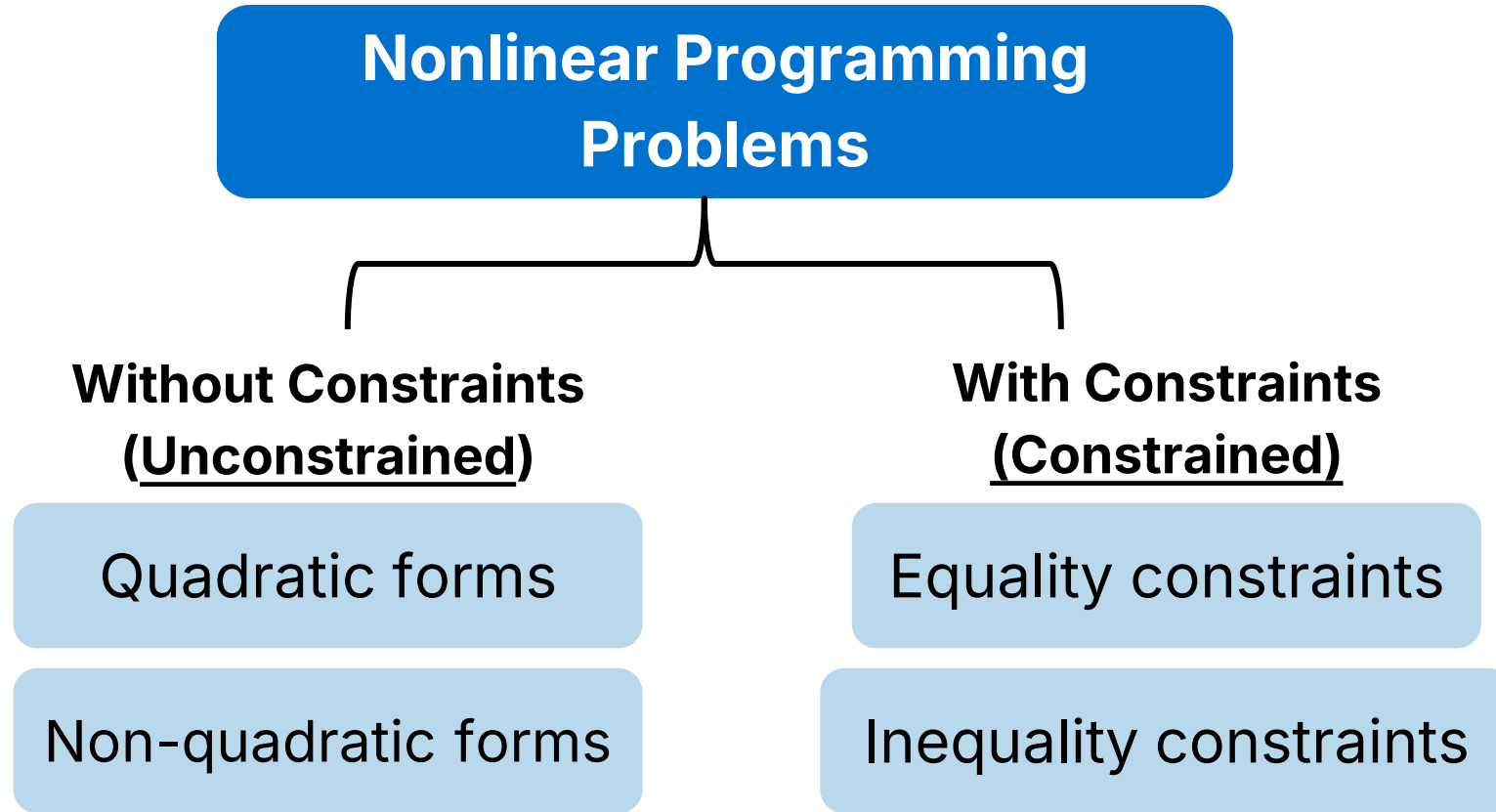
Without constraints

Non-Quadratic form

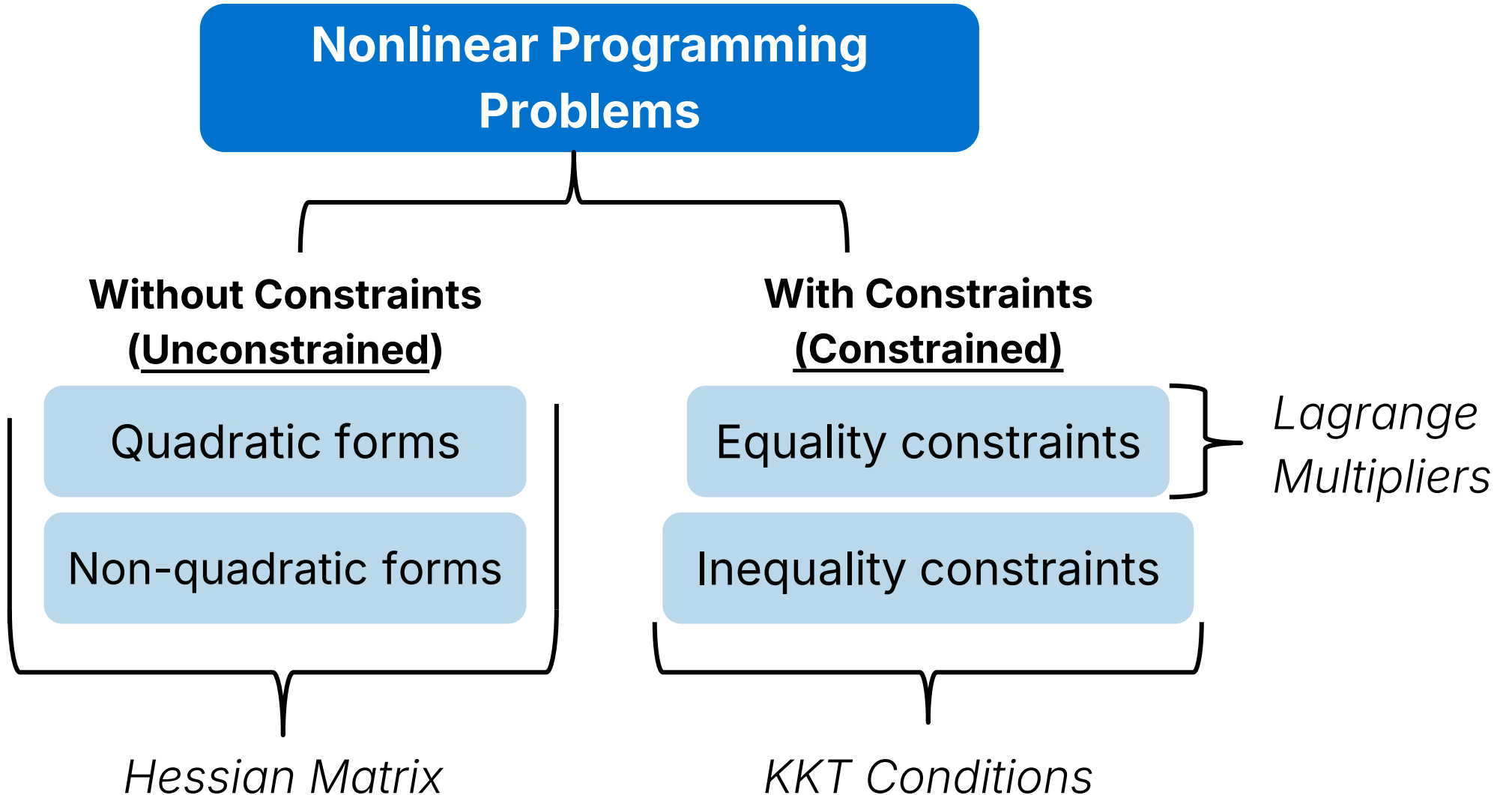
Types of NLPs



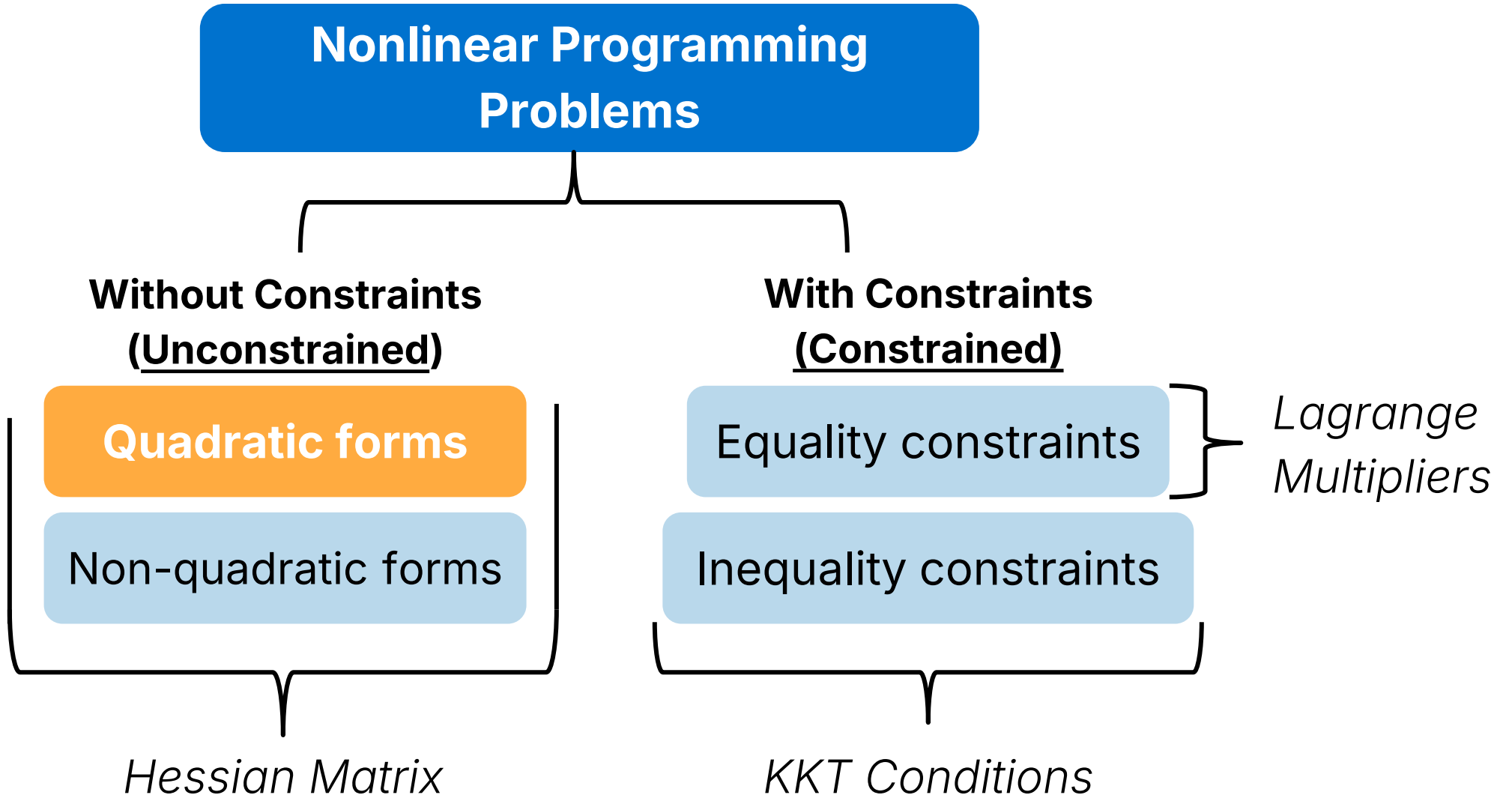
Types of NLPs



Types of NLPs



Types of NLPs

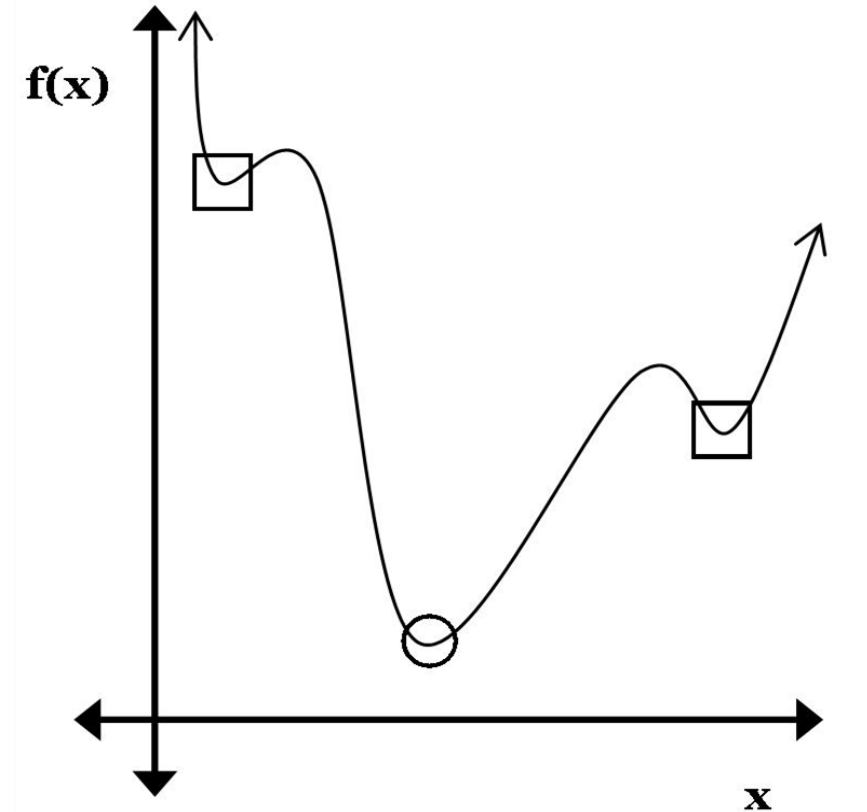


Convexity

Convexity | Local vs Global Optima

For continuous optimization, the “easy” problems are those where **local** optimality implies **global** optimality.

- Improving search algorithms for such problems will not get stuck at the wrong local optimum
- We would like conditions on the objective function and the constraints that guarantee this.
- If the function is convex, a local minimum is also a global minimum

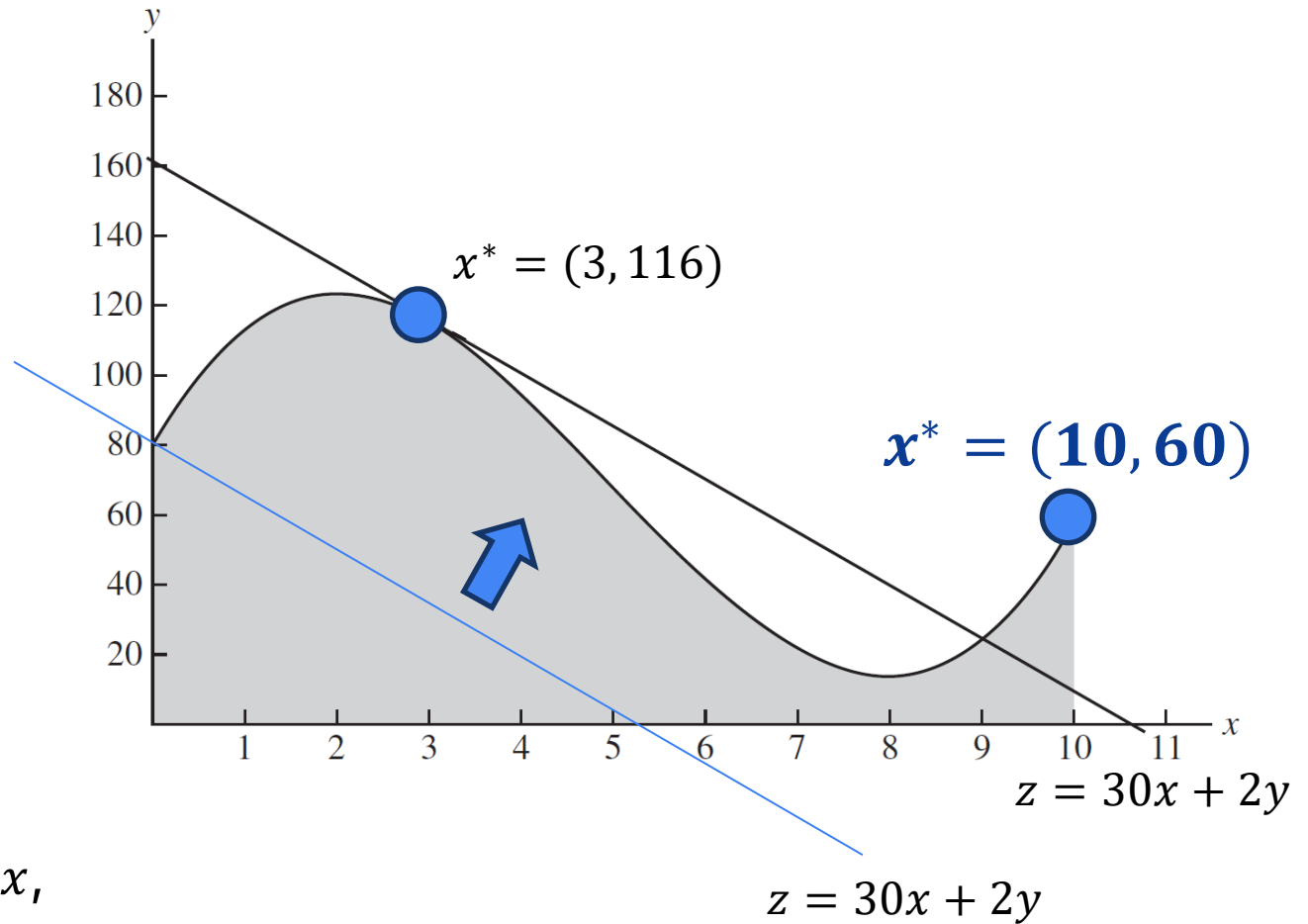


Convexity | Local vs Global Optima

Consider this problem:

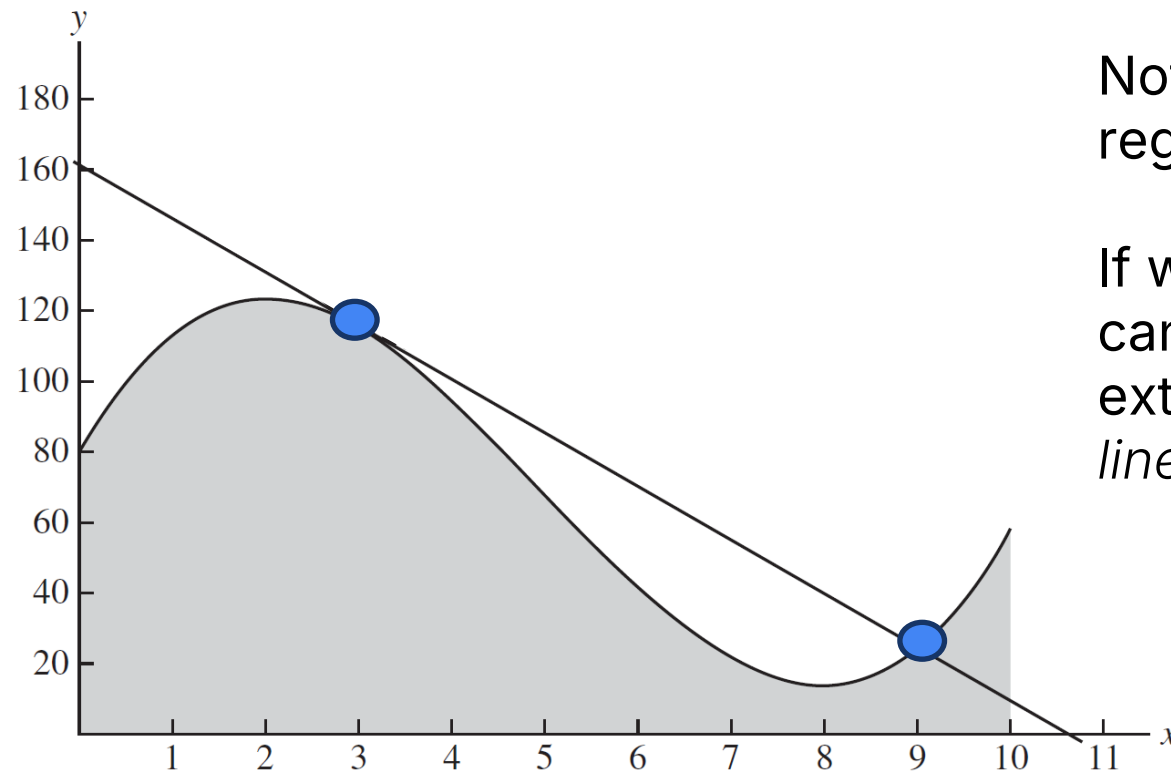
$$\begin{aligned} \max \quad & z = 30x + 2y \\ \text{s.t.} \quad & y \leq x^3 - 15x^2 + 48x + 80 \\ & x \leq 10 \\ & x, y \geq 0. \end{aligned}$$

The local maximum arises because the first constraint, viewed as a function of x , changes from concave to convex!



Convexity | Local vs Global Optima

Can we describe when a local maximum can occur in a more general way?



Notice that the line left the feasible region at $x = 3$ but reentered it at $x = 9$

If we only consider regions where lines cannot do this, the problem of local extrema will be eliminated (*at least for linear objective functions!*)

This is exactly how convexity of a feasible region will be defined.

Convexity | Local vs Global Optima

The same condition on the feasible region works for nonlinear objectives if they have the correct shape (concavity).

Let's change the objective in previous example:

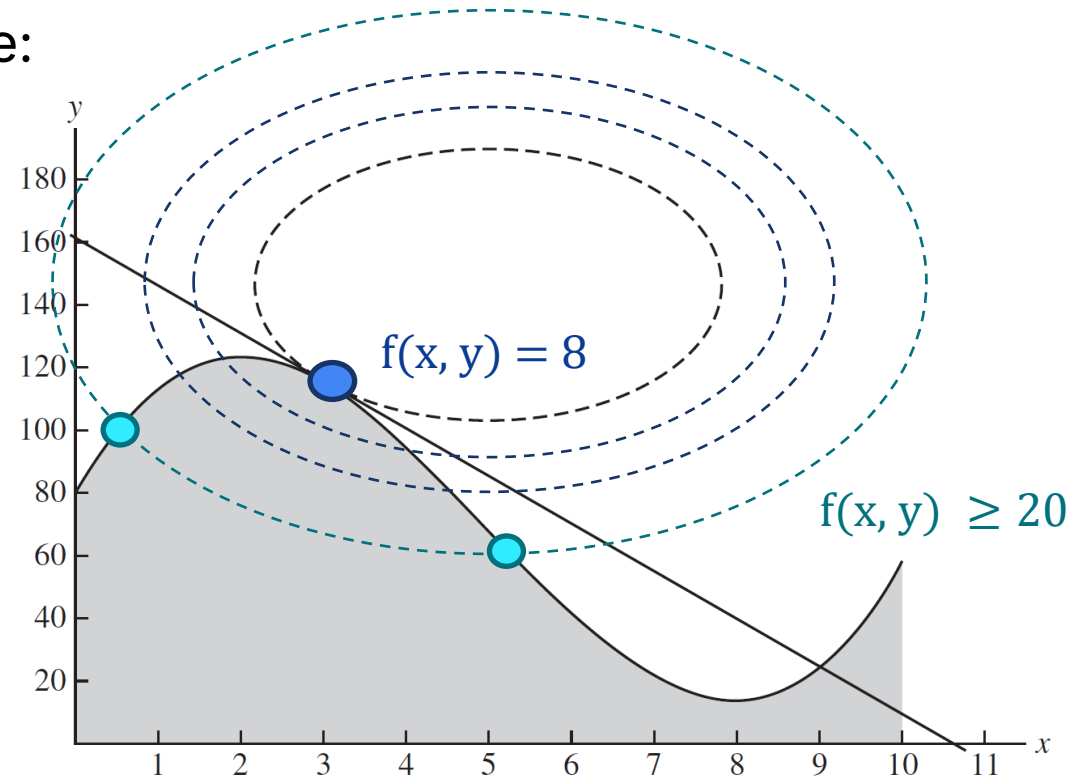
$$\min f(x, y) = (x - 5)^2 + \frac{1}{225} (y - 146)^2$$

$$\text{s.t. } y \leq x^3 - 15x^2 + 48x + 80$$

$$x \leq 10$$

$$x, y \geq 0.$$

The entire region inside the ellipse, where $f(x, y) \leq 8$, lies above tangent line



Convex Sets (Feasible Regions)

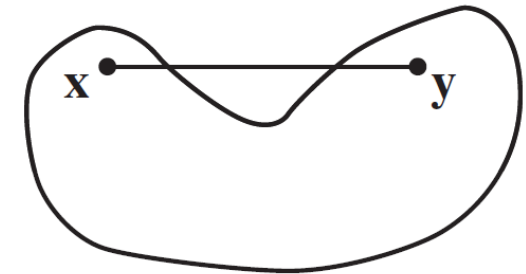
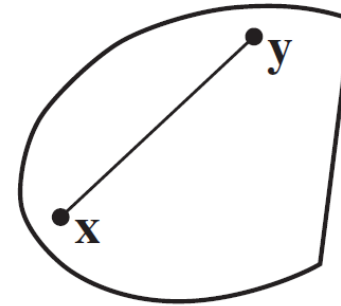
The set $S \in \mathbb{R}^n$ is **convex** if for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$, $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} \in S$.

For example, if

$\mathbf{x} = (7, 0, -2)$ and $\mathbf{y} = (1, 3, 4)$,

the line segment between them can be

$$\lambda \begin{bmatrix} 7 \\ 0 \\ -2 \end{bmatrix} + (1 - \lambda) \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 6 \\ -3 \\ -6 \end{bmatrix}$$



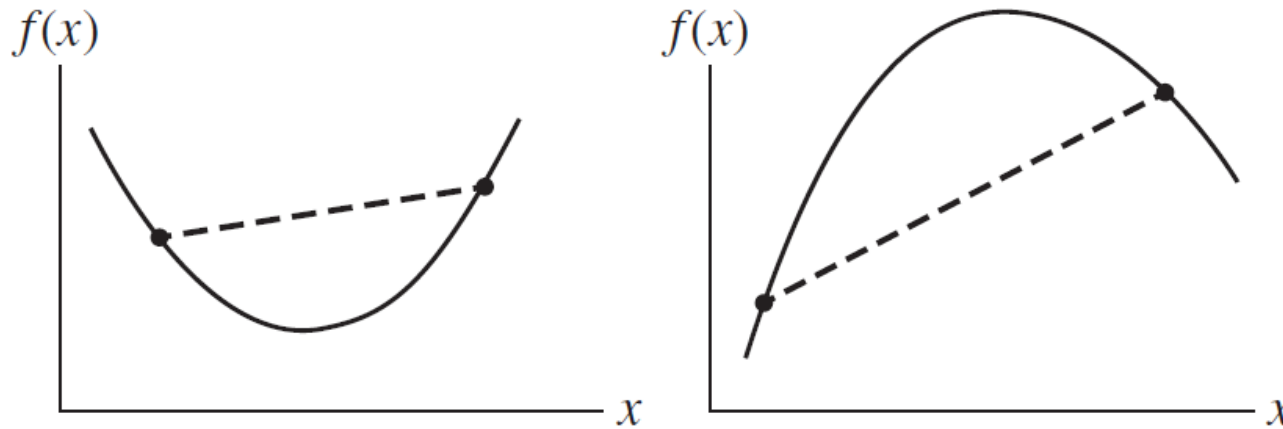
Choosing $\lambda = \frac{1}{3}$,

$$(1 + 2, 3 - 1, 4 - 2) = (3, 2, 2),$$

one-third of the distance from $\mathbf{x}$ to $\mathbf{y}$.

Convex and Concave Functions

A function is **convex** if all secant lines* lie above (or on) the function and **concave** if all secant lines lie below (or on) the function.



For unconstrained optimization problems, convexity of the objective function is sufficient to guarantee that a **local minimum is also a global minimum**.

**a straight line that intersects a curve at two or more distinct points*

Convex and Concave Functions

A function f on a convex set S is **convex** if for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$,
$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

A function f on a convex set S is **concave** if for all $\mathbf{x}, \mathbf{y} \in S$ and $\lambda \in [0, 1]$,
$$f(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \geq \lambda f(\mathbf{x}) + (1 - \lambda) f(\mathbf{y}).$$

*Many functions are convex on some interval and concave on another;
such functions are neither convex nor concave!*

Example | Convex and Concave Functions

Can you show that $g(x_1, x_2) = x_1^2 + x_2$ is convex?

For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}$ and $\lambda \in [0, 1]$, we must show that:

$$[\lambda x_1 + (1 - \lambda)y_1]^2 + \lambda x_2 + (1 - \lambda)y_2 \leq \lambda(x_1^2 + x_2) + (1 - \lambda)(y_1^2 + y_2).$$

Then,

$$\begin{aligned} \lambda(x_1^2 + x_2) + (1 - \lambda)(y_1^2 + y_2) - \lambda(1 - \lambda)x_1^2 + 2\lambda(1 - \lambda)x_1y_1 - \lambda(1 - \lambda)y_1^2 \\ \leq \lambda x_1^2 + \lambda x_2 + (1 - \lambda)y_1^2 + (1 - \lambda)y_2, \end{aligned}$$

which simplifies to:

$$-x_1^2 + 2x_1y_1 - y_1^2 = -(x_1 - y_1)^2 \leq 0$$

Convex and Concave Functions

For constrained problems, convexity of a feasible region can be checked by checking the convexity/ concavity of its inequality constraints!

Let S be a convex set.

If f is convex and $\mathbf{x}^$ is a local minimum of $\min_{\mathbf{x} \in S} f(\mathbf{x})$ then $\mathbf{x}^*$ is optimal.*

If f is concave and $\mathbf{x}^$ is a local maximum of $\max_{\mathbf{x} \in S} f(\mathbf{x})$ then $\mathbf{x}^*$ is optimal.*

Convex Programming

$$\max f(\mathbf{x})$$

$$\text{s.t. } g_i(\mathbf{x}) \leq b_i, \quad i = 1, \dots, m$$

is a **convex program** if f is concave and all g_i are convex.

Example:

$$\max \quad z = -(x - 2)^2 - (y - 1)^2$$

$$\text{s.t. } x + y \leq 4$$

$$x^2 + y^2 \leq 9$$

$$x, y \geq 0$$

Can you visualize this feasible region & objective contour lines?

Convex Programming

$$\min f(\mathbf{x})$$

$$\text{s.t. } g_i(\mathbf{x}) \geq b_i, \quad i = 1, \dots, m$$

For a minimization problem to be a convex program, f must be convex; for constraints $g_i(\mathbf{x}) \geq b_i$, g_i must be concave.

Example:

$$\min f(x, y) = (x - 1)^2 + (y - 2)^2$$

$$\text{s.t. } -x^2 - y^2 + 4 \geq 0$$

$$x + y \geq 1$$

Can you visualize this feasible region?

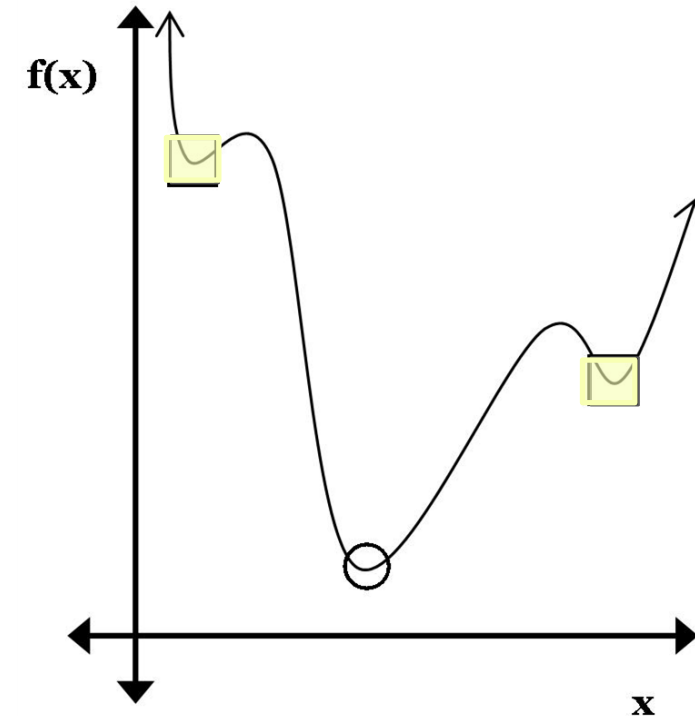
Challenges in NLPs | Local vs. Global Optima

One of the challenges of nonlinear optimization is that, in general, there may be many **local optima**

x is a local minimum of $f(\cdot)$ if $f(x) \leq f(y)$
for every feasible y sufficiently close to x

That is, $x_j - \epsilon \leq y_j \leq x_j + \epsilon$
for all j and some small ϵ

Although local optima satisfy condition that the first derivative equals zero, they are, in general, of no interest



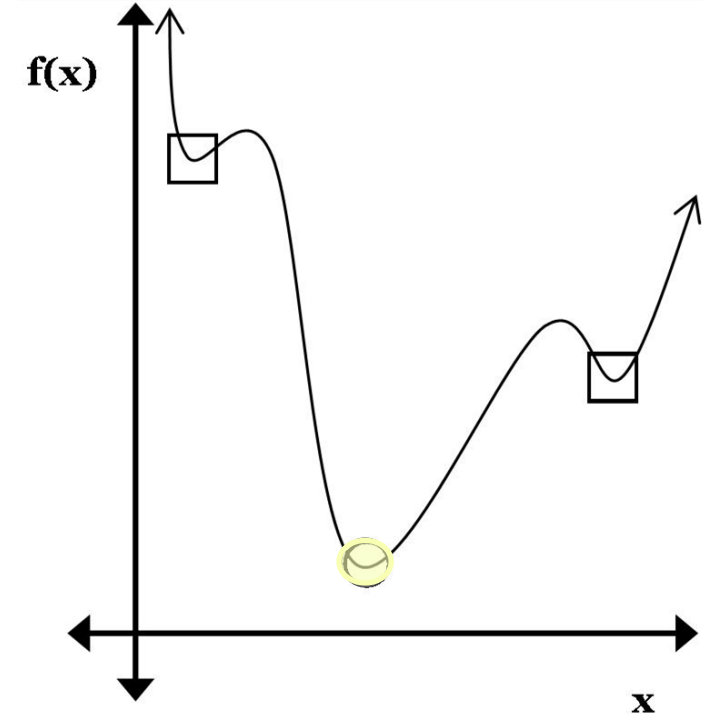
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Although local optima satisfy condition that the first derivative equals zero, they are, in general, of no interest



Ideally, we want to find a **global optimum**

x is a global minimum of $f(\cdot)$ if $f(x) \leq f(y)$ for every feasible y

Quadratic Programming

- Introduction to Quadratic Programming
- Quadratic Programming Using Gurobi

Quadratic Programming (QP)

Quadratic Form:

$$f(x) = c_{11}x_1^2 + c_{22}x_2^2 + \cdots + c_{nn}x_n^2 + c_{12}x_1x_2 + \cdots + c_{\{n-1,n\}}x_{n-1}x_n = x^T A x$$

Where $x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$ and $A = (a_{ij})_{\{n \times n\}}$ is a square matrix

Also, $a_{\{ii\}} = c_{\{ii\}}$ and $a_{\{ij\}} = a_{\{ji\}} = \frac{c_{ij}}{2}, i \neq j$

Quadratic Programming (QP)

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Sometimes we might work with **QPs** that are in this form too:

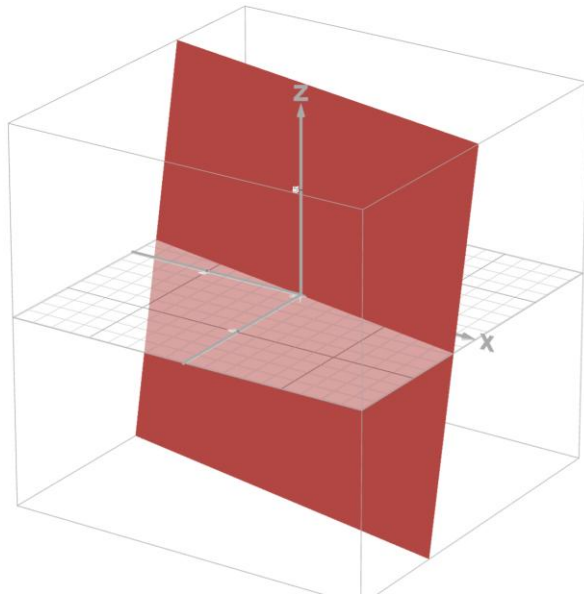
$$\frac{1}{2}x^T Q x + R^T x + S$$

Where $x = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}$ and $A = (a_{ij})_{\{n \times n\}}$ is a square matrix

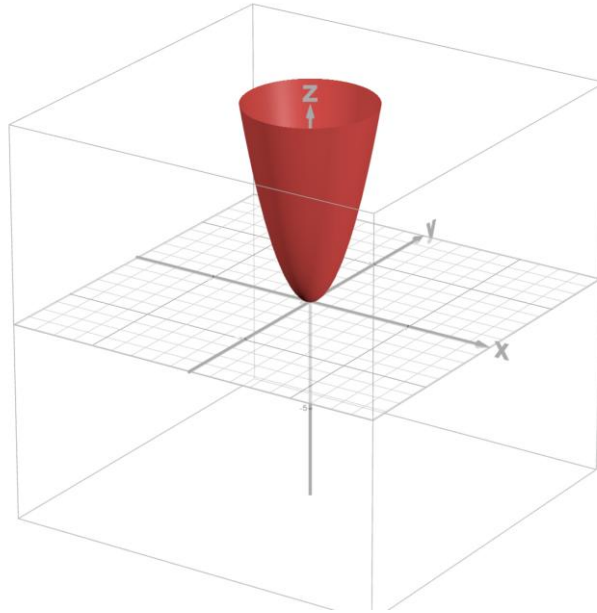
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Quadratic Programming (QP)

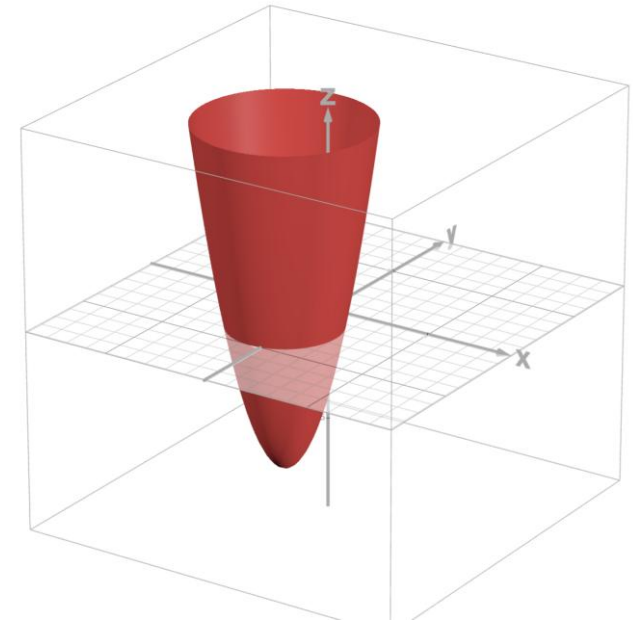
$$Z_1 = x + 5y$$



$$Z_2 = x^2 + y^2$$



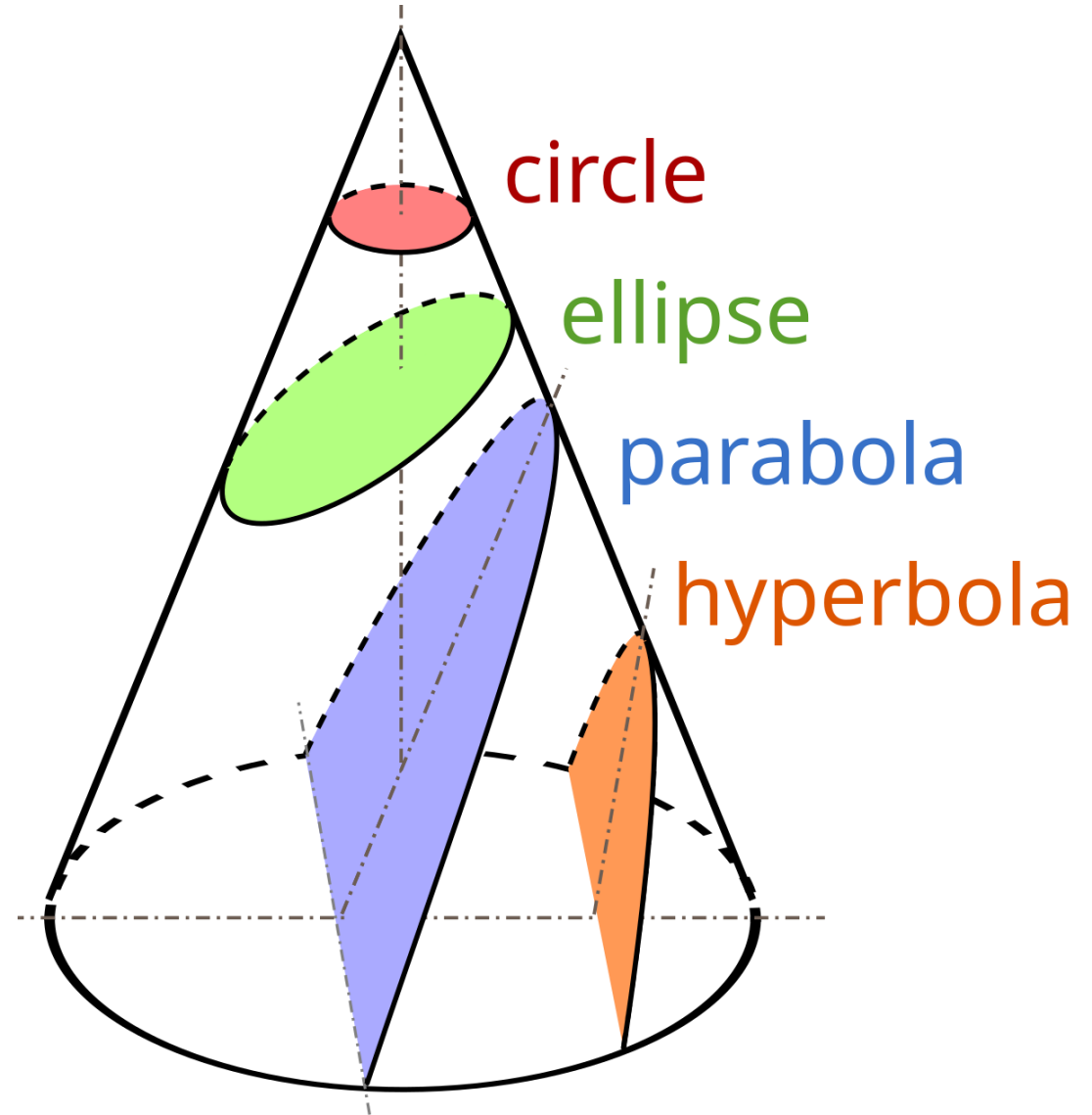
$$Z_3 = x^2 + y^2 + x + 5y$$



Quadratic Programming (QP)

Quadratic function describe
a conic section

Conic Section	Equation
Circle	$x^2 + y^2 = a^2$
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
Parabola	$y^2 = 4ax$
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



Quadratic Programming (QP)

Quadratic Form:

$$f(x) = c_{11}x_1^2 + c_{22}x_2^2 + \cdots + c_{nn}x_n^2 + c_{12}x_1x_2 + \cdots + c_{\{n-1,n\}}x_{n-1}x_n = x^T A x$$

Where $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $A = (a_{ij})_{\{n \times n\}}$ is a square matrix

Also, $a_{\{ii\}} = c_{\{ii\}}$ and $a_{\{ij\}} = a_{\{ji\}} = \frac{c_{ij}}{2}, i \neq j$

Here A is a symmetric matrix that contains the coefficients that define the curvature of the function.

Matrix A And Convexity

A can be used to show (strict) convexity of function f :

- A is positive definite if and only if $x^T A x > 0$ for all $x \neq 0$ in domain of f .
- If A is **positive definite**, then f is **convex**!

Ways to show that A is positive definite:

1. Use *successive principal minors*: Start at the top left corner, and show the (1×1) , (2×2) , ... $(n \times n)$ (sub)matrix of A has a positive determinant.
2. Show that **all** eigenvalues of A are positive.
3. Any solver will gladly tell you, as well (!)...

QP Example

Given $f(X) = x_1^2 + 2x_2^2 - 7x_3^2 - 4x_1x_2 + 6x_1x_3 - 5x_2x_3$

Could you find the coefficient matrix A ?
Consider the relationship between A and the quadratic form $f(X)$ being $X^TAX = f(X)$

QP Example

Given $f(X) = x_1^2 + 2x_2^2 - 7x_3^2 - 4x_1x_2 + 6x_1x_3 - 5x_2x_3$
 $= X^TAX$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 2 & -5/2 \\ 3 & -5/2 & -7 \end{bmatrix}$$

The relationship between A and the quadratic form $f(X)$ is:

$$X^TAX = f(X)$$

```
At python command prompt:  
import numpy as np  
m2 = np.matrix([[1, -2, 3],[-2,2,-5/2],[3,-5/2,-7]])  
print(np.linalg.eigvals(m2))
```

QP Example

Given $f(X) = x_1^2 + 2x_2^2 - 7x_3^2 - 4x_1x_2 + 6x_1x_3 - 5x_2x_3$
 $= X^T A X$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & -2 & 3 \\ -2 & 2 & -5/2 \\ 3 & -5/2 & -7 \end{bmatrix}$$

**Not
convex!**

The relationship between A and the quadratic form $f(X)$ is:

$$X^T A X = f(X)$$

Matrix A , Eigenvalues And Convexity

Eigenvalues of A	Shape of the Function
All eigenvalues of A are positive	$f(\mathbf{x})$ is strictly convex
All eigenvalues of A are non-negative	$f(\mathbf{x})$ is convex, which may not be strict
All eigenvalues of A are negative	$f(\mathbf{x})$ is strictly concave
All eigenvalues of A are non-positive	$f(\mathbf{x})$ is concave, which may not be strict

Quadratic Programming Using Gurobi

QP: An Application In Manufacturing

Suppose the *Firerock* company produces tires by combining three ingredients: rubber, oil, and carbon black

The costs for each are:

- Rubber: \$0.04 per pound
- Oil: \$0.01 per pound
- Carbon black: \$0.07 per pound

Determine how to combine ingredients

- Do so at minimum cost
- However, design constraints exist...



Firerock Company Example

To manufacture a set of four automobile tires:

- Need exactly 100 pounds of total product
- Must have at least 50 pounds of carbon black
- Must have between 25 and 60 pounds of rubber



Additionally, several physical attributes must be satisfied:

- **Hardness** of no more than 35
- **Elasticity** of at least 16
- **Tensile strength** of at least 12

Hardness: $34 + 0.10r + 0.06o - 0.3c +$

$$0.002r^2 + 0.005o^2 + 0.001c^2 + 0.001r * o$$

Elasticity: $17 + 0.35r - 0.04o - 0.002o^2$

Tensile Strength: $12.5 - 0.10o - 0.001 o^2$

Firerock Company Example

Begin with variable definitions to produce four tires:

r: Pounds of rubber in mixture

o: Pounds of oil in the mixture

c: Pounds of carbon black in the mixture

Statistical analysis shows that the hardness, elasticity, and tensile strength of a 100-pound mixture of rubber, oil, and carbon black is:

Hardness: $34 + 0.10r + 0.06o - 0.3c + 0.002r^2 + 0.005o^2 + 0.001c^2 + 0.001r * o$

Elasticity: $17 + 0.35r - 0.04o - 0.002o^2$

Tensile Strength: $12.5 - 0.10o - 0.001 o^2$

Firerock Company Example

Begin with variable definitions to produce four tires:

r: Pounds of rubber in mixture

o: Pounds of oil in the mixture

c: Pounds of carbon black in the mixture

Statistical analysis shows that the hardness, elasticity, and tensile strength of a 100-pound mixture of rubber, oil, and carbon black is:

Hardness: $34 + 0.10r + 0.06o - 0.3c + 0.002r^2 + 0.005o^2 + 0.001c^2 + 0.001r * o$

Elasticity: $17 + 0.35r - 0.04o - 0.002o^2$

Tensile Strength: $12.5 - 0.10o - 0.001 o^2$

This is a quadratically constrained problem (**QCP**) with a linear objective!

Solving the Firerock Example using Gurobi

```
\This is the firerock.lp file format for the rubber tire problem

minimize
0.04 r + 0.01 o + 0.07 c

subject to
r + o + c = 100
c >= 50
r >= 25
r <= 60
\Tensile strength
[ - 0.001 o ^2 ] - 0.1 o >= -0.5
\Elasticity
[ - 0.002 o ^2 ] + 0.35 r - 0.04 o >= -1
\Hardness
[ 0.002 r ^2 + 0.001 r * o + 0.005 o ^2 + 0.001 c ^2 ] + 0.1 r + 0.06 o - 0.3 c <= 1
end
```

Solution to Problem

Reading problem into Gurobi and solving yields:

- **What is all of that output?**

Uses an *interior point / barrier algorithm* to perform optimization

- **Optimal solution uses:**

- $r^* = 45.23$ pounds of rubber

- $o^* = 4.77$ pounds of oil

- $c^* = 50.00$ pounds of carbon black

- **Minimum cost is about \$5.36**

Important!

Remember that Gurobi can solve **convex, quadratically constrained** optimization problems with quadratic objective functions

- If **minimization**, objective function should be **convex**
- If **maximization**, objective function should be **concave**

Example | Smith Communications

Smith Communications provides cellular phone service. They want to expand to provide intercity service between four cities in northern Ohio. A new communications tower must be built to handle these inter-city calls. The tower will have a 40-mile transmission radius. Here are the locations, in (x, y) coordinates, for the four cities:

- Cleveland (5,45)
- Akron (12,21)
- Canton (18,5)
- Youngstown (52, 21)

Example | Smith Communications

Formulate an optimization problem that will determine the optimal placement of the new tower that minimizes the collective distance to the four cities.

- Be sure to ensure that each city is still within the 40-mile transmission radius.
- Use standard Euclidean distance (i.e., ℓ_2 norm) to compute the distances, where the objective is to sum each of the squared distances

Example | Smith Communications

The distance to any city (x_1, y_1) can be computed using the ℓ_2 -norm as)

and its square as:

$$\sqrt{(x - x_1)^2 + (y - y_1)^2}$$

$$(x - x_1)^2 + (y - y_1)^2$$

Then summing the squares of these distances, we have:

Minimize:

$$(x - 5)^2 + (y - 45)^2 + (x - 12)^2 + (y - 21)^2 + (x - 18)^2 + (y - 5)^2 + (x - 52)^2 + (y - 21)^2$$

Subject to:

$$\begin{aligned} (A) \quad & (x - 5)^2 + (y - 45)^2 \leq 40^2, \\ (B) \quad & (x - 12)^2 + (y - 21)^2 \leq 40^2, \\ (C) \quad & (x - 18)^2 + (y - 5)^2 \leq 40^2, \\ (D) \quad & (x - 52)^2 + (y - 21)^2 \leq 40^2. \end{aligned}$$

Gurobi Solution

- Read the cell_tower.lp file
- Note that the objective function and constraints were formed by squaring terms and recombining.
- The optimal placement of the new tower is $x^* = 21.75$, and $y^* = 23$